Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Mathematics II**

Code No: **MTH168**   Semester: **2nd Semester**

Full Marks: **60**   Pass Marks: **24**   Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Define homogeneous linear system of equations with an example. Which type of homogeneous equation has a nontrivial solution? Determine the value of \(x, y \), and \(z \) if the system of equation \(x - 2y + 3z = 0 \), \(-x + 2y - 4z = 0 \), \(2x - 4y + 9z = 0 \) has a nontrivial solution.

2. Let \(T \) be defined by \(T(x) = Ax \), where \(A = \begin{bmatrix} 1 & -5 & -7 \\ -3 & 7 & 5 \end{bmatrix} \). Find a vector \(x \) whose image under \(T \) is \(b \), where \(b = \begin{bmatrix} -2 \\ -2 \end{bmatrix} \), and determine whether \(x \) is unique or not.

3. (a) Find the basis and dimension of the subspace \(H = \left\{ \begin{bmatrix} a - 3b + 6c \\ 5a + 4d \\ b - 2c - d \\ 5d \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\} \).\[[1+4] \](b) What is rank of a matrix? Find the rank of a matrix \(A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix} \).

Section B

Attempt any Eight questions[8×5=40]

4. Let \(A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} \). Find \(A^3 \).

5. Let \(A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} \). Find \(5|A| - 2|I| \). Is det(5A) equal to 5 det \(A^2 \) Justify.

6. Define subspace of a vector space. Prove that \(H = \left\{ \begin{bmatrix} a \\ 0 \\ c \end{bmatrix} : a, c \in \mathbb{R} \right\} \) is a subspace of \(\mathbb{R}^3\).

7. Find the eigenvalues for the given matrix \(A = \begin{bmatrix} 4 & 0 & 1 \\ -1 & -6 & -2 \\ 5 & 0 & 0 \end{bmatrix} \).

8. Let \(x = \begin{bmatrix} 7 \\ 6 \end{bmatrix} \) and \(u = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \). Find the orthogonal projection of \(x \) onto \(u \). Also, write \(x \) as the sum of two orthogonal vectors, one in span \(\{u\}\) and one orthogonal to \(u \).

9. Find LU factorization of \(A = \begin{bmatrix} 3 & 2 \\ -2 & 3 \end{bmatrix} \).

10. Use Cramer's rule to solve the equations \(\frac{4}{x} + \frac{6}{y} = 4 \) and \(\frac{3}{x} - \frac{4}{y} = \frac{1}{6} \).

11. Does $(\mathbb{Z},.)$ form a group? Justify.

12. If R is a ring with additive identity 0, then prove that for any $a, b \in R$(i) $0a = a0 = 0$(ii) $(-a)(-b) = ab$.